# 1 Powers of sine and cosine

1. Evaluate $\int \cos^2(x)dx$. (*Hint:* $\cos^2(x) = \frac{1}{2}(1+\cos(2x))$.

2. Evaluate $\int \cos^4(x)dx$.

3. Evaluate $\int \sin^2(x)dx$. (*Hint:* $\sin^2(x) = \frac{1}{2}(1-\cos(2x))$.

4. Evaluate $\int \sin^4(x)dx$.

5. Evaluate $\int \cos^3(x)dx$. (*Hint*: $\cos^3(x) = \cos^2(x)\cos(x) = (1-\sin^2(x))\cos(x)$.)

6. Evaluate $\int \cos^5(x)dx$

7. Evaluate $\int \sin^3(x)dx$. (*Hint*: $\sin^3(x) = \sin^2(x)\sin(x) = (1-\cos^2(x))\sin(x)$.)

8. Evaluate $\int \sin^5(x)dx$

9. Explain in your own words how to calculate $\int \cos^n(x)dx$ or $\int \sin^n(x)dx$ when $n$ is a positive integer.

## 2 $\int \sin^m(x)\cos^n(x)dx$

10. Evaluate $\int \sin^n(x)\cos(x)dx$ where $n$ is a positive integer.

11. Evaluate $\int \sin^n(x)\cos^3(x)dx$ where $n$ is a positive integer.

12. Evaluate $\int \sin^3(x)\cos^n(x)dx$ where $n$ is a positive integer.

13. Evaluate $\int \sin^2(x)\cos^2(x)dx$. (*Hint*: Use the double-angle identities $\sin^2(x) = \frac{1}{2}(1-\cos(2x))$ and $\cos^2(x) = \frac{1}{2}(1+\cos(2x))$.)

14. Explain in your own words how to calculate $\int \sin^m(x)\cos^n(x)dx$ when $m$ and $n$ are positive integers.

## 3 $\int \sin(mx)\sin(nx)dx$, $\int \cos(mx)\cos(nx)dx$, and $\int \sin(mx)\cos(nx)dx$

For these integrals, we use the following trig identities:

(a) $\sin(\alpha)\sin(\beta) = \frac{1}{2}(\cos(\alpha-\beta)-\cos(\alpha+\beta))$

(b) $\cos(\alpha)\cos(\beta) = \frac{1}{2}(\cos(\alpha-\beta)+\cos(\alpha+\beta))$

(c) $\sin(\alpha)\cos(\beta) = \frac{1}{2}(\sin(\alpha-\beta)+\sin(\alpha+\beta))$

15. Evaluate $\int \sin(mx)\sin(nx)dx$ when $m$ and $n$ are integers.

16. Evaluate $\int \cos(mx)\cos(nx)dx$ when $m$ and $n$ are integers.

17. Evaluate $\int \sin(mx)\cos(nx)dx$ when $m$ and $n$ are integers.